

A note on mixed tree coloring

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Abstract

In this note we observe that the problem of mixed graph coloring can be solved in linear time for trees, which improves the quadratic algorithm of Hansen et al. [P. Hansen, J. Kuplinsky, D. de Werra, Mixed graph colorings, *Math. Methods Oper. Res.* 45 (1997) 145–160].

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1. Introduction

Let $G = (V, E)$ be a simple graph of order $|V| = n$, and let $\odot$ be an orientation of some subset of its edges, $\odot \subseteq \{(i, j) \in V \times V : \{i, j\} \in E\}$, which does not induce directed circuits. Then $G_\odot = (V, E, \odot)$ is called a *mixed graph*. A k -coloring of $G_\odot$ is a function $\varphi : V \rightarrow \{1, 2, \dots, k\}$ such that $\varphi(i) \neq \varphi(j)$ for $\{i, j\} \in E$ and $\varphi(i) < \varphi(j)$ for $(i, j) \in \odot$. The smallest k for which $G_\odot$ has a k -coloring is called the *chromatic number* of mixed graph $G_\odot$ and denoted by $\chi_\odot(G)$.

Whereas classical graph coloring may be thought of as the problem of solving a given set of inequalities of the form $\varphi_i \neq \varphi_j$ with variables $\{\varphi_1, \dots, \varphi_n\}$ in the range of values $\{1, 2, \dots, k\}$, mixed graph coloring solves a more general set of inequalities, in which additional conditions of the form $\varphi_i < \varphi_j$ are allowed. Such

relations often appear in scheduling problems where both incompatibility requirements and precedence constraints are present. Sotskov et al. [3] showed applications of the model in job-shop scheduling for the problem $J|p_i = 1|C_{\max}$, some other fields of application are mentioned in [1,4].

The chromatic number $\chi_\odot(G)$ of a mixed graph can be bounded in terms of the classical chromatic number of G and the number of vertices $L_\odot(G)$ of the longest directed path in the orientation $\odot$ [1]. In particular, when G is bipartite, there are only two possible values of $\chi_\odot(G)$:

$$L_\odot(G) \leq \chi_\odot(G) \leq L_\odot(G) + 1. \quad (1)$$

Deciding which of these two bounds is tight for a given bipartite graph is known to be NP-hard [2]. For the case of trees, Hansen et al. [1] proposed an $O(n^2)$ algorithm based upon the concept of 0-, 1-, 2-edges. Herein we provide a much simpler recursive algorithm, with linear time complexity.

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2. Outline of approach

Let $T = (V, E, \odot, r)$ be a mixed tree with root r . For a given vertex $v \in V$, let $T(v)$ denote the mixed subtree of T having root v and induced by the set of all vertices $u \in V$ such that v lies on the path connecting u and r . In subtree $T(v)$, define $X(v)$ as the set of neighbors of v such that for any $x \in X(v)$ we have $(x, v) \in \odot$, let $Y(v)$ be the set of neighbors of v such that for any $y \in Y(v)$ we have $(v, y) \in \odot$, and let $Z(v)$ be the set of all the remaining neighbors of v .

The notation $[a, b]$ is used to describe the set of all integers contained between a and b inclusive. Suppose we want to obtain a coloring of the tree using at most k colors. We define function $\Phi : V \rightarrow 2^{[1, k]}$ which associates with each vertex v the set $\Phi(v)$ of all possible color values which may be assigned to v in any k -coloring of $T(v)$; if no such coloring exists, $\Phi(v) = \emptyset$. Observe that by a simple induction argument, function Φ has the following recursive (bottom-up) characterization:

- (1) If $v \in V$ is a leaf, i.e. $X(v) = Y(v) = Z(v) = \emptyset$, then $\Phi(v) = [1, k]$.
- (2) Suppose that $v \in V$ is not a leaf. If in $T(v)$ there exists a vertex $u \in X(v) \cup Y(v) \cup Z(v)$ such that $\Phi(u) = \emptyset$, then $\Phi(v) = \emptyset$. Otherwise, for any integer $\varphi \in [1, k]$ we have $\varphi \in \Phi(v)$ if and only if for each vertex $u \in X(v) \cup Y(v) \cup Z(v)$ there exists an integer $\varphi(u) \in \Phi(u)$ such that the following conditions are fulfilled:
 - (a) $\varphi \geq \varphi(u) + 1$, for all $u \in X(v)$;
 - (b) $\varphi \leq \varphi(u) - 1$, for all $u \in Y(v)$;
 - (c) $\varphi \neq \varphi(u)$, for all $u \in Z(v)$.

If for all $v \in V$ we have $\Phi(v) \neq \emptyset$, then one can easily construct a proper coloring of tree T : for the root we arbitrarily choose a color from the set $\Phi(r)$, and then in a top-down manner we extend the partial coloring to the subtree of an already colored vertex—this is always possible by clause (2) of the characterization of function Φ .

The proposed algorithm is based upon the observation that for each vertex v , instead of computing the whole set $\Phi(v)$, it is possible to find its minimal and maximal elements, only.

More precisely, consider the function $\Phi^* : V \rightarrow \{[a, b] : 1 \leq a \leq b \leq k\} \cup \emptyset$, recursively defined as follows:

- (1) If $v \in V$ is a leaf, then let $\Phi^*(v) = [1, k]$.
- (2) Suppose that $v \in V$ is not a leaf. If in $T(v)$ there exists a vertex $u \in X(v) \cup Y(v) \cup Z(v)$ such that

$\Phi^*(u) = \emptyset$, then let $\Phi^*(v) = \emptyset$. Otherwise, let $a \in [1, k]$ be the minimum possible integer such that:

- (a) $a \geq \max_{x \in X(v)} \min \Phi^*(x) + 1$;
- (b) $\Phi^*(z) \neq [a, a]$, for all $z \in Z(v)$.

Likewise, let $b \in [1, k]$ be the maximum possible integer such that:

- (c) $b \leq \min_{y \in Y(v)} \max \Phi^*(y) - 1$;
- (d) $\Phi^*(z) \neq [b, b]$, for all $z \in Z(v)$.

Then, if such a and b exist and $a \leq b$, we set $\Phi^*(v) = [a, b]$, otherwise we set $\Phi^*(v) = \emptyset$.

Proposition 1. Let $T = (V, E, \odot, r)$ be a mixed tree. Then for every vertex $v \in V$ we have

$$\Phi^*(v) = \begin{cases} \emptyset, & \text{if } \Phi(v) = \emptyset, \\ [\min \Phi(v), \max \Phi(v)], & \text{otherwise.} \end{cases} \quad (2)$$

Proof. The proof proceeds by induction with respect to the order of the subtree.

If v is a leaf, then clearly $\Phi(v) = \Phi^*(v) = [1, k]$. Now, suppose that the induction hypothesis holds for all vertices in $T(v)$ apart from v , in particular, for all neighbors of v from sets $X(v)$, $Y(v)$, and $Z(v)$. The implication $\Phi(v) = \emptyset \Rightarrow \Phi^*(v) = \emptyset$ is clear; we confine ourselves to the remaining case $\Phi(v) \neq \emptyset$, hence we can assume $\Phi(u) \neq \emptyset$ for all $u \in X(v) \cup Y(v) \cup Z(v)$.

From the characterization of the function Φ , we see that an integer $\varphi \in [1, k]$ belongs to $\Phi(v)$ if and only if the following conditions are fulfilled:

- (i) $\varphi \geq \max_{x \in X(v)} \min \Phi(x) + 1$;
- (ii) $\varphi \leq \min_{y \in Y(v)} \max \Phi(y) - 1$;
- (iii) $\Phi(z) \neq \{\varphi\}$, for all $z \in Z(v)$.

Directly from the induction hypothesis we obtain a set of equivalent conditions, that is, $\varphi \in \Phi(v)$ if and only if the following conditions are fulfilled:

- (i') $\varphi \geq \max_{x \in X(v)} \min \Phi^*(x) + 1$;
- (ii') $\varphi \leq \min_{y \in Y(v)} \max \Phi^*(y) - 1$;
- (iii') $\Phi^*(z) \neq [\varphi, \varphi]$, for all $z \in Z(v)$.

Let $a = \min \Phi^*(v)$ and $b = \max \Phi^*(v)$. Comparing conditions (i') and (iii') with clauses (a) and (b) in the definition of function Φ^* , it is clear that limit a can be equivalently defined as the minimum integer in interval $[1, k]$ which satisfies conditions (i') and (iii'). Since conditions (i') and (iii') are necessarily fulfilled by all elements of $\Phi(v)$, we have $a \leq \min \Phi(v)$. Likewise, b is the maximum integer which satisfies conditions (ii') and (iii'), hence $b \geq \max \Phi(v)$. However, as $\Phi(v) \neq \emptyset$ by assumption, we see that a must also

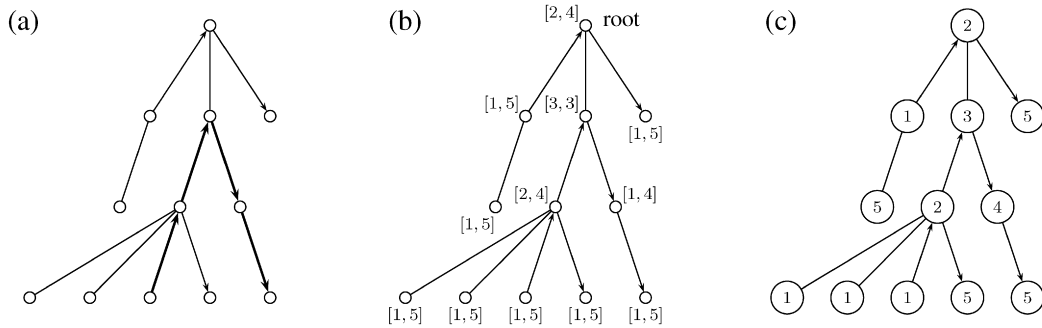


Fig. 1. (a) A mixed tree T , and the longest directed path. (b) The computed function Φ^* for $k=5$. (c) An optimal 5-coloring of T .

Input: A mixed tree $T = (V, E, \mathbb{O}, r)$.

Output: An optimal coloring of T (and thus also the value of $\chi_{\mathbb{O}}(T)$).

1. Determine the maximal length of a directed path in the orientation, $L_{\mathbb{O}}(T)$.
2. Compute the intervals describing the function Φ^* with $k = L_{\mathbb{O}}(T)$. If for each vertex $v \in V$, $\Phi^*(v) \neq \emptyset$, then go to Step 4.
3. Compute the intervals describing the function Φ^* with $k = L_{\mathbb{O}}(T) + 1$.
 /* Notice that by relation (1), there exists a k -coloring of T with $k = L_{\mathbb{O}}(T) + 1$, thus after this step, for each vertex $v \in V$ we have $\Phi^*(v) \neq \emptyset$. */
4. Determine a k -coloring φ of T using the top-down approach:
 - 4.1. Set $\varphi(r) = a$, where $\Phi^*(r) = [a, \cdot]$.
 - 4.2. Given an already colored vertex v , extend the partial coloring to each of its neighbors u in subtree $T(v)$:
 - 4.2.1. If $(v, u) \in \mathbb{O}$, then set $\varphi(u) = c$, where $[\cdot, c] = \Phi^*(u)$.
 - 4.2.2. If $(u, v) \in \mathbb{O}$, then set $\varphi(u) = c$, where $[c, \cdot] = \Phi^*(u)$.
 - 4.2.3. Else, if $\Phi^*(u) = [c, c]$, then we set $\varphi(u) = c$. If $\Phi^*(u) = [a, b]$ and $a \neq b$, then we can choose $\varphi(u) \in \{a, b\}$ so that $\varphi(u) \neq \varphi(v)$.

Algorithm 1. Optimal coloring of a mixed tree in linear time.

satisfy condition (ii), while b must also satisfy condition (i), so $a, b \in \Phi(v)$. This leads to the conclusion that $\min \Phi(v) = a = \min \Phi^*(v)$ and $\max \Phi(v) = b = \max \Phi^*(v)$, which completes the proof. $\square$

3. The algorithm

Taking into account relations (1) and (2), we are ready to formulate Algorithm 1. The running time of

Step 1 is linear with respect to the number of vertices. Next, it is easy to observe that for a vertex $v \in V$, $\Phi^*(v)$ can be computed in $O(\deg(v))$ time, where $\deg(v)$ is the number of neighbors of v . Thus Step 2 (resp., Step 3) can be implemented in a $\sum_{v \in V} O(\deg(v)) = O(n)$ running time, since T is a tree. Finally, the optimal mixed k -coloring φ of T is also constructed in $O(n)$ time in such a way that each node v is assigned one of the two permissible color values $\{\min \Phi^*(v), \max \Phi^*(v)\} = \{\min \Phi(v), \max \Phi(v)\}$ (note that in general $\Phi^*(v) \neq \Phi(v)$).

Theorem 2. An optimal coloring of a mixed tree can be determined in linear time.

Fig. 1 illustrates an exemplary execution of Algorithm 1 for a tree T with $L_{\mathbb{O}}(T) = \chi_{\mathbb{O}}(T) = 5$.

References

- [1] P. Hansen, J. Kuplinsky, D. de Werra, Mixed graph colorings, *Math. Methods Oper. Res.* 45 (1997) 145–160.
- [2] B. Ries, Coloring some classes of mixed graphs, *Discrete Appl. Math.* 155 (1) (2007) 1–6.
- [3] Y.N. Sotskov, A. Dolgui, F. Werner, Mixed graph coloring for unit-time job-shop scheduling, *Int. J. Math. Algorithms* 2 (4) (2001) 289–323.
- [4] Y.N. Sotskov, V.S. Tanaev, F. Werner, Scheduling problems and mixed graph colorings, *Optimization* 51 (3) (2002) 597–624.